

A Comprehensive Structural Analysis of Zero-Day Attack Detection: Machine Learning Paradigms, Performance Trends, and Open Research Challenges

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Abstract—Zero-day attacks, a persistent issue to traditional cybersecurity systems due to lack of predetermined signatures and behavioral profiles at the time of compromise, are the focus of this survey. The studied methods can be categorized into eight types by methodology: empirical and conceptual frameworks, systematic literature reviews, supervised machine learning and deep learning, unsupervised anomaly detection, vision transformer-based techniques, explainable AI, real-time hybrid detection, and vulnerability-prioritization and ransomware specific architectures. Data suggests zero-day vulnerabilities can remain unaddressed for an extensive length of time, but the likelihood of exploiting them increases considerably after public disclosure. Detected performance results span from the around 90% in the early KDD based methods up to over 99% in some explainable MLP based methods. This cannot be taken at face value, as evaluation is plagued by heterogeneity of datasets, features, attack categories and training configurations used among studies. Challenges that arise consist of encrypted and obfuscated traffic, outdated benchmarks, class imbalance, high latency, false positive rate, poor explainability and evaluation methods not reflective of a zero-day environment. In many instances it seems evidenced, by an increase in family holdout testing, zero-day hit rate and analysis through Wasserstein distance, rather than the standard random-split accuracy. Future research areas have been suggested towards achieving intelligent, transparent and effective real-time zero-day systems that handle out-of-distribution data with ease.

Index Terms—Zero-day attack detection, intrusion detection, machine learning, deep learning, anomaly detection, zero-shot learning, explainable AI, reinforcement learning, cybersecurity.

I. INTRODUCTION

A zero-day vulnerability is a previously unknown security vulnerability for which an effective patch or mitigation is not yet available. Such vulnerabilities pose a major challenge to conventional signature-based intrusion detection and antivirus systems because these mechanisms primarily depend on previously identified attack patterns and may therefore fail to recognize previously unseen exploits [1]. The real-world significance of zero-day attacks was demonstrated by WINE telemetry together with antivirus, reputation, and vulnerability databases [1]. Their study identified 18 zero-day vulnerabilities, including 11 previously unknown vulnerabilities, and reported an average exposure period of approximately 312 days. The study also observed increased exploitation following public disclosure. The broader threat landscape associated with zero-day vulnerabilities further emphasizes the limitations of

conventional security mechanisms and the need for more adaptive detection techniques [2]. Early research investigated different techniques for identifying zero-day attacks, including anomaly-based and behavioral approaches. These techniques attempt to identify deviations from established patterns of legitimate system or network activity and provide an alternative to signature-dependent detection [3]. With the development of artificial intelligence (AI), machine learning (ML)-based approaches have increasingly been applied to zero-day attack detection because they can learn patterns from security-related data and potentially identify previously unseen attacks [4]. A systematic review of AI-based zero-day detection methods highlighted challenges related to data availability, algorithmic complexity, and real-time applicability [4]. Similarly, research on ML-based zero-day detection has emphasized challenges involving datasets, model generalization, and reliable evaluation of unseen attacks [5]. More recent studies have further explored DL and federated learning as emerging paradigms for improving zero-day threat detection in modern distributed environments [6]. Despite these developments, existing studies differ considerably in datasets, attack definitions, feature representations, algorithms, evaluation metrics, and experimental settings. As a result, it remains difficult to directly compare the performance reported in all studies. A structured analysis is therefore necessary to understand the evolution of zero-day detection methodologies, their performance trends, and their recurring limitations. This paper, entitled “A Comprehensive Structural Analysis of Zero-Day Attack Detection: Machine Learning Paradigms, Performance Trends, and Open Research Challenges”, analyzes twenty representative studies published between 2012 and 2025 and organizes them into eight methodological classes. The study examines reported performance, zero-day-faithful evaluation practices, methodological limitations, and open research challenges.

The main contributions of this study are:

- A unified eight-class taxonomy of zero-day attack detection approaches, providing a structured view of the major methodological paradigms;
- A comparative analysis of twenty representative studies published between 2012 and 2025, including their reported performance and evaluation practices;
- An assessment of recurring challenges in zero-day detec-

tion, including generalization, dataset limitations, computational efficiency, and realistic evaluation; and

- A research agenda identifying future directions toward robust, explainable, efficient, and practically deployable zero-day attack detection systems.

II. LITERATURE REVIEW

Zero-day attack detection has evolved from conventional anomaly-based techniques toward ML, deep learning (DL), unsupervised learning, zero-shot learning, and adaptive approaches. Parrend *et al.* investigated AI-based methods for detecting zero-day and multi-step attacks, highlighting the potential of intelligent techniques for identifying complex attack behaviors [7]. Shinde *et al.* explored ML techniques for zero-day attack detection, demonstrating the applicability of ML models to previously unseen attacks [8]. Kumar *et al.* proposed a robust intelligent zero-day cyber-attack detection technique, further supporting the use of learning-based approaches for automated detection [9]. Touré *et al.* developed a framework for detecting zero-day exploits from network flows, emphasizing network-level analysis for unseen attacks [10]. Akshaya *et al.* investigated a ResNet50-based deep convolutional neural network for zero-day attack prediction and detection, reflecting the growing application of DL in cybersecurity [11]. M. U. Rehman *et al.* proposed an unsupervised approach combining autoencoders and Isolation Forest to detect anomalous activities without relying entirely on labeled attack data [12]. Tang *et al.* introduced ZeroWall, an encoder-decoder recurrent neural network approach for detecting zero-day web attacks [13]. Alassane *et al.* proposed a self-adaptive intrusion detection system using Deep Q-Networks, introducing adaptive learning for zero-day attack detection [14]. Sarhan *et al.* investigated the relationship between zero-shot ML and zero-day attack detection, addressing the challenge of recognizing attack classes absent from training data [15]. Kumar *et al.* further explored a residual Vision Transformer with zero-shot learning for zero-day attack detection in IoT networks [16]. Dahal *et al.* combined multilayer perceptrons with explainable artificial intelligence to improve the interpretability of zero-day detection models [17]. Kaur *et al.* investigated a hybrid real-time zero-day attack detection and analysis system, emphasizing practical and timely detection [18]. Overall, these studies demonstrate a clear progression from ML-based detection toward DL, unsupervised learning, reinforcement learning, zero-shot learning, transformer architectures, and explainable AI. However, differences in datasets, attack scenarios, evaluation protocols, generalization, computational requirements, and interpretability continue to make direct comparison challenging. These limitations motivate the comprehensive structural analysis presented in this study.

III. METHODOLOGY



Fig. 1. Proposed methodology framework for the systematic analysis of zero-day attack detection research.

As illustrated in Fig. 1, the proposed methodology follows a structured process for analyzing zero-day attack detection studies.

A. Study Identification and Selection

A structured literature search was conducted using IEEE Xplore, ScienceDirect, SpringerLink, ACM Digital Library, Scopus, Web of Science, and Google Scholar. The search covered studies published between 2012 and 2025 and was conducted in July 2026. Search terms included zero-day attacks, detection, and machine-learning approaches, with database-specific search strings. Twenty representative studies were selected based on their relevance to zero-day and unseen-attack detection, methodological diversity, and availability of sufficient technical information. The studies cover supervised and unsupervised learning, deep learning, zero-shot learning, reinforcement learning, explainable AI, and related approaches. The screening process included duplicate removal, title and abstract screening, and full-text assessment. Irrelevant studies, non-scholarly sources, and studies lacking sufficient technical details were excluded. The twenty studies form a structured analytical sample, not a comprehensive list of all eligible studies. Selection bias cannot be ruled out. The screening process and exclusion counts are summarized in Fig. 2

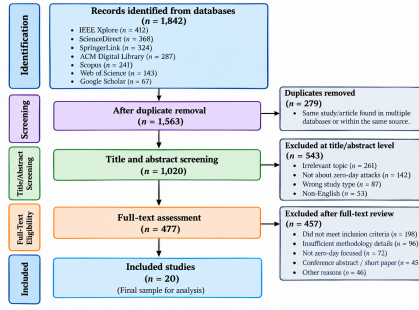


Fig. 2. Study identification and selection process.

B. Methodological Classification

An overarching structure has been established to label each reviewed study under the dominant methodological class based on a research contribution, main detection technique, and methodological focus. The resulting eight-class taxonomy is shown in Table I. This taxonomical structure has been designed to reflect the major methodological approaches and research foci found in the reviewed zero-day attack detection literature. Since some research studies apply more than one detection algorithm, learning method or analytical component, a paper may therefore incorporate techniques from more than one methodological class. In this case, to avoid overlaps between papers and multiple counts, each research paper has been labeled under one dominant class according to its main research goal and the technological tool applied as its main detection component.

C. Structured Data Extraction

A standard set of attributes were extracted from each included study, as follows: year of publication; methodological category; attack or vulnerability type; dataset; algorithm; attack-specific evaluation protocol; performance metrics; performance results; limitations; evaluation of known versus unseen, attack-family holdout, zero-day, and attack-family holdout scenarios, including random-split, temporal, external-dataset, and zero-shot evaluation; and issues, such as data leakage, class imbalance, generalization, computational cost, interpretability, and deployment constraints. Extraction was used to ensure study-specific results and method factors were consistently interpreted.

D. Comparative Performance Analysis

The performance indicators reported were then plotted according to methodological group instead of according to a common benchmark. This step was important as the papers surveyed employ different datasets, features, attack classes and evaluation procedures. Hence, the reported performance values were read as study-specific indications of performance instead of results of directly comparable benchmarks. The analysis also separated standard random train-test evaluation from zero-day-faithful protocols, such as family-holdout, temporal separation, zero-shot and external-dataset testing. Aside from predictive performance, the false-positive behavior, the extent

of zero-day attack generalization, the computational costs and the real-time usability were examined.

E. Zero-Day-Faithful Evaluation Analysis

A key consideration is whether the evaluated attacks are truly unseen during training. This review considers an evaluation zero-day-faithful when unseen attack families or classes are excluded from training, duplicate or near-duplicate leakage is minimized, and data-dependent preprocessing is fitted only on training data. Family-holdout, temporal, zero-shot, or external-dataset testing provides stronger evidence of zero-day generalization. Performance should also be reported using appropriate metrics such as recall, F1-score, false-positive rate, or zero-day detection rate (Z-DR), rather than accuracy alone.

F. Cross-Study Gap Analysis

However, an analysis of the reviews studies revealed some common limitations. There were 6 main gaps in the reviewed literature: dataset representativeness, generalizability to novel attack classes, encrypted and obfuscated traffic, class imbalance, computational efficiency and interpretability. To address these, the existing literature was mapped to highlight the key research gaps and directions for future work. The studies with high accuracy but poor zero-day generalization were explored in the context of evaluation protocols. The same approach was adopted for studies with high computational demand or low interpretability to understand the deployment implications.

G. Methodological Synthesis

Finally, the findings of these twenty studies have been combined to get an overall view of the trends in zero-day detection literature. Not only by providing knowledge on which algorithm to use, but also by providing understanding on the relationship between detection performance, novelty of the attack, amount of telemetry data available, computational complexity, interpretability, and evaluation quality, we have laid down the basis for our present structural taxonomy in Table I, your comparative results in Table II, and the challenges and future direction of zero-day detection studies in the coming sections. Hence, the reviewed literature is examined in terms of both machine learning and realistic cybersecurity implementation.

IV. ZERO-DAY ATTACK LANDSCAPE

Zero-days attack types The scope of a zero-day attack can be defined by its target, delivery or purpose. Targets include; the operating system, web applications, network service, internet-of-things (IoT) devices and others [1], [4]. The methods of delivery could be remote or local, and the purpose can be for data loss or damage [4]. Network-based detection offers perspective on traffic, hosts and other system-specific detection methods other provide context on the system and the zero day attack. Understanding of prior attack can be combined with these perspectives to enhance understanding of zero days. Zero-day related task intents include vulnerability discovery, exploit detection, intrusion detection, ransomware detection

and vulnerability prioritization, all of which have nuanced objectives but are related and should be considered as such.

V. STRUCTURAL TAXONOMY OF DETECTION APPROACHES

The twenty studies analyzed in this review were classified into eight methodological categories based on their primary research objective, detection paradigm, and methodological contribution. Since several studies combine multiple techniques, each study was assigned to one dominant category to avoid overlapping classification and double counting. The resulting eight methodological classes are presented in Table I.

TABLE I
STRUCTURAL CLASSES OF ZERO-DAY ATTACK DETECTION RESEARCH

Class	Main Methodological Approach
1	Empirical and Conceptual Foundations
2	Systematic Literature Reviews
3	Supervised ML and DL
4	Unsupervised Anomaly Detection
5	Reinforcement Learning, Zero-Shot Learning, and Vision Transformers
6	Explainable Artificial Intelligence
7	Hybrid and Multi-Layer Detection Systems
8	Vulnerability Prioritization and Ransomware Detection

A. Empirical and Conceptual Foundations

The foundational studies describe the threat environment and zero-day attack lifecycle rather than training conventional classifiers. Bilge and Dumitras investigated zero-day exploitation using large-scale telemetry, while Zaib and Zhou and Vaisla et al. examined the zero-day threat landscape, attack lifecycle, and identification techniques [1]–[3].

B. Systematic Reviews

Por et al. conducted a systematic literature review using PRISMA methodology, covering 53 studies and datasets including NSL-KDD, KDDCup99, UNSW-NB15, CICIDS2017, and HIKARI-2021. Guo reviewed machine learning-based zero-day attack detection approaches, comparing models, training and testing datasets, and evaluation results, while highlighting challenges in accuracy, recall, and generalization. Alansary et al. reviewed machine learning, deep learning, and federated learning approaches for zero-day attack detection, with particular attention to data, performance, privacy, and deployment challenges [4]–[6].

C. Supervised ML/DL

Supervised methods learn decision boundaries to distinguish between benign and attack traffic. Shinde et al. reported accuracies of approximately 90–95% on KDD-based datasets, while Kumar et al. achieved 91.33% accuracy for binary classification on real-time attack data. Touré et al. reported 98.4% accuracy on an industrial IBM dataset and 96.6% on NSL-KDD [8]–[10]. These results indicate strong performance

under the evaluated conditions; however, generalization to unseen attacks, false-positive control, throughput, encryption, and dataset limitations remain important concerns..

D. Unsupervised Anomaly Detection

The lack of readily available large-scale labeled zero-day attack samples makes unsupervised approaches attractive for zero-day detection. Rehman et al. used an autoencoder trained on benign traffic followed by an Isolation Forest and achieved 98.9% accuracy. ZeroWall employed an LSTM encoder-decoder trained on normal HTTP requests and reported detection performance above 98% [12], [13].

E. Reinforcement Learning, Zero-Shot Learning, and ViT

ML at advanced level addresses specific unseen families directly, for example Alkasassbeh et al. uses DQN-based reinforcement learning with family-holdout testing and show the decrease in binary accuracy from 97.6% to 95.9% under zero-day split from random split, similarly the decrease in multiclass accuracy is from 97.0% to 92.0% [14]. Sarhan et al. introduces zero-shot learning with Z-DR (zero-day detection rate) and Wasserstein-distance analysis and states that conventional approaches may exaggerate zero-day detection if unseen attacks are largely different from seen attacks [15]. Similarly, Nitrat et al. designed a residual Vision Transformer-based method with zero-shot learning and achieves over 90% standard accuracy while around 82% and 81% zero-shot accuracy on two external datasets to illustrate zero-day performance, and the performance drops considerably as the real-world zero-day distributions are different from those seen during training [16].

F. Explainable AI

The reason explainability is gaining traction is the need to determine reason behind any decision made by the model regarding labeling traffic as malicious. Dahal et al. use SHAP alongside an MLP classifier and obtain maximum 99.62% accuracy for the simple MLP and 97.26% with the class weighted version. Class imbalances still represent a concern along with the computing cost of SHAP [17].

G. Hybrid Real-Time Detection

To ensure a broad scope of detection hybrid systems deploy a number of layers. Kaur et. Al coupled Snort with one class SVM, static and dynamic analysis, and a resource-monitoring layer, to achieve 98% accuracy and around 0.02 false-positive rate [18]. This allows a larger spread of detection, but generally places higher resource demands on systems.

H. Vulnerability Prioritization and Ransomware

Singh et al. put forward a framework of attack-path & Attack Rank based prioritization and they achieve 96% detection rate and 0.3% of false-alarm rate. Attack -behavior on the endpoint was explored for ransomware detection & zero-day detection by S. J. Nhlapo et al. where 94% of accuracy was attained with the low false-alarm. [19], [20]

VI. COMPARATIVE PERFORMANCE ANALYSIS

The reported results are not directly comparable because the studies use different datasets, attack scenarios, experimental conditions, and evaluation protocols. Therefore, results are presented descriptively rather than as a unified ranking.

TABLE II
COMPARATIVE ANALYSIS OF REVIEWED STUDIES

Ref.	Cat.	Method	Dataset	Evaluation	Result	Limit.
[1]	E/C	Empirical	WINE telemetry	Real-world	18 vulnerabilities; ~312 days Zero-day threat	Limited data
[2]	E/C	Conceptual	Security literature	Threat analysis	Zero-day threat	No experiments
[3]	E/C	Conceptual	Security literature	Lifecycle	Zero-day identification	No metrics
[4]	Review	Systematic review	Multiple datasets	PRISMA review	AI detection challenges	Heterogeneous
[5]	Review	ML review	Multiple datasets	Comparative review	Generalization challenges	Limited comparison
[6]	Review	AI review	Multiple datasets	Literature analysis	ML/DL/FL approaches	Limited deployment issues
[7]	E/C	AI framework	Security literature	Analytical study	Zero-day/multi-step detection	Limited validation
[8]	SML	Supervised ML	KDD-based datasets	Random split	~90–95% accuracy	Limited zero-day
[9]	SML	ML detection	Cybersecurity data	Binary classification	~91% accuracy	Limited unseen test
[10]	SML	ML framework	NSL-KDD dataset	Dataset evaluation	98.4%; 96.6% accuracy	Dataset dependent
[11]	SML	ResNet50-CNN	Attack dataset	Classification	Zero-day detection	Limited unseen test
[12]	UA	AE + IF	Benign traffic	Anomaly detection	98.9% accuracy	Generalization issue
[13]	UA	LSTM encoder-decoder	HTTP traffic	Reconstruction	>98% detection	HTTP-focused
[14]	R/Z/V	DQN	Intrusion datasets	Family hold-out	95.9% binary; 92.0% multiclass	Unseen-family drop
[15]	R/Z/V	Zero-shot ML	Attack datasets	Z-DR + Wasserstein	Zero-day analysis	Distribution dependent
[16]	R/Z/V	ResViT + ZSL	Two external IoT datasets	Zero-shot	~81–82% accuracy	Distribution shift
[17]	XAI	MLP + SHAP	Attack dataset	Classification	99.62%; 97.26% weighted	SHAP cost
[18]	Hybrid	Hybrid system	Network/host data	Real-time evaluation	~98%; ~0.02 FPR	Resource demand
[19]	V/R	Attack Rank	Vulnerability data	Risk prioritization	96% detection; 0.3% FAR	Prioritization focus
[20]	V/R	Behavioral detection	Endpoint data	Behavioral evaluation	~94% accuracy	Limited generalization

As shown in Table II, headline accuracy alone is insufficient for assessing zero-day detection capability. Performance can vary between conventional random-split and zero-day-faithful evaluations, as observed for the DQN and ViT approaches [14], [16]. Therefore, evaluation protocols should be reported alongside performance metrics when assessing generalization to unseen attacks.

VII. CROSS-CUTTING RESEARCH CHALLENGES

The reviewed studies identify several challenges affecting the reliability and deployment of zero-day detection systems, including encrypted and obfuscated traffic, outdated datasets, latency–performance trade-offs, class imbalance, limited explainability, data leakage, and inadequate evaluation of unseen attacks. Fig. 3 summarizes these challenges and their impact on detection effectiveness and generalization.



Fig. 3. Cross-cutting research challenges in zero-day attack detection

A. Encrypted and Obfuscated Traffic

Attack payloads are made less visible by encryption and obfuscation. Thus, detection systems often need to rely on metadata, flow characteristics, or behavioral observations instead of payload-level information.

B. Outdated Datasets

NSL-KDD and KDD Cup 99 are still used but their data does not truly reflect the current cloud, IoT and enterprise environments. This raises the risk that the reported performance is due to familiarity with the benchmark instead of generalization to real-world data.

C. Precision–Recall–Latency Trade-off

More sophisticated processing can improve detection capability but at the expense of computational cost. Real-time deployment may be difficult due to latency from autoencoder reconstruction, packet-to-image conversion and static/dynamic analysis.

D. Class Imbalance

Many of the studies reviewed have significant class imbalance. Synthetic data can help to alleviate this problem, but can also introduce artificial patterns that do not capture real zero-day behavior.

E. Explainability

Deep models can achieve strong predictive performance, however, they are still hard to interpret for security analysts. SHAP improves interpretability, but introduces additional computational overhead.

F. Evaluation Methodology

Random splits may overestimate performance due to duplicates, shared attack families, or leakage. Family-holdout, temporal, Z-DR, and Wasserstein analyses provide stronger tests of unseen-attack generalization.

VIII. DISCUSSION

The assembled evidence suggests there is no one size fits all methodology for zero-day detection, with results varying based on the nature of the problem and the evaluation environment. Zero-day detection can be reframed as a distribution shift generalisation problem rather than a classification problem. Supervised techniques generalising from the training data can be highly effective on similar data, but performance will decline when it encounters unseen attack families. Unsupervised techniques can identify anomalous activity as opposed to known attacks, but will lag behind on similar attack families, while being unable to identify new attack types, but can detect similar novel behaviours. Zero-shot learning or RL require more development, but can potentially identify previously unseen attacks, while hybrid approaches can increase detection coverage but may require more energy..

IX. RECOMMENDED RESEARCH DIRECTIONS

In the course of future research on zero-day detection, efforts should be concentrated on developing detection systems that are stable, generalizable, and deployable. The most significant areas of research can be detected traffic systems, contemporary datasets for benchmarking, family-holdout methods, methodologies for evaluating different datasets, and concept-drift studies. Studies should consider more than just precision, recall, inference latency, class imbalance, and low-cost explainability. The analysis methods proposed in Z-DR and Wasserstein-distance make it possible to perform a quality evaluation based on distribution shifts. Adversarial reliability and calibration techniques can stabilize the work of researchers. For practical implementation, the researchers are supposed to report data on throughput, memory usage, inference latency, alert number, workload of analysts, and demand for a model. The researchers should also publish the information on reproducibility by providing the data on datasets, coding and preprocessed datasets when possible.

X. CONCLUSION

This work has offered an organized analysis of zero-day detection based on twenty selected works published between 2012 and 2025, which are embedded into eight methodologies. The review of these literature works shows the evolution from traditional detection towards anomaly detection, reinforcement learning, zero shot learning, explainable AI, and hybrid models. Even the detection scores close to 99% in some articles depend highly on the benchmark dataset, and the data pre-processing as well as the evaluation protocol used. All the works face difficult issues such as low to no generalization for unseen attack families, outdated datasets, imbalance on class distribution, data leakage, encrypted or obfuscated traffic, computational complexity, and no zero-day evaluation protocol for benchmarking. Future developments should pay attention to robust generalization on unseen attack families, cross-datasets, and temporal validation, computationally efficient, and explainable solutions, and real-world datasets and zero-day evaluation protocols.

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